

Installation Standards- Informal

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I ABBREVIATIONS, ACRONYMS, AND DEFINITIONS

DR	Drop
PON	Passive optical network
AC	Alternating Current
DC	Direct Current
UPS	Uninterrupted power supply
FTN	Fibertime Networks
ONT	Optical Network Terminal
ODN	Optical distribution network

II RELEVANT AND APPLICABLE DOCUMENTS

Document No	Description



1. Version History

Version	Author	Date	Comments
0.1	Jared Honey	21/07/2022	Initial release
0.2	Jared Honey	25/07/2022	Addition of mesh AP information
0.3	Godfrey Krige	04/05/2023	Addition of Mini UPS Power Solution
0.4	Johann Small	13/11/2023	Addition of formal house installation

2. Objective

This document is to be used for reference when installing the consumer equipment connecting to a network.

3. Target Audience

This document is intended for fibertime™ staff and all contractors involved in the deployment of a fibertime™ network.

4. Introduction

The installation standards have been developed by fibertime™ to all contractor's flexibility while installing consumer equipment into the consumer's home. The effort that has gone into creating this document is the result of input and collaboration from fibertime™ partners. This document should be used as a guideline within all fibertime™ networks.

- This document will cover the following aspects of the installation:
 - The entry of the fibre into a home.
 - The routing of cables, both electrical and optical.
 - The assembly of the ONT mounting bracket.
 - The mounting the ONT bracket.
 - The mounting of the Mini UPS Power Solution.

5. Installation Standards

5.1 Entry

The entry into the home should be selected using the following criteria:

- Approximately the midpoint of the house.
- Entry should be into a common area within the house such as a living room.
- The entry should take into consideration any tenants that are not living in



the house structure. The entry point may be into a non-common area such as a bedroom should this be a more suitable location to provide tenants with Wi-Fi coverage.

- Should the entry point be into a non-common area, the customer must be made aware of the limitations of the Wi-Fi coverage. Factors like the thickness of walls and distance to other areas within the house must be noted to the customer should the entry point be in a non-common area.
- The entry point should be within a 5m cable run of an electrical outlet within the house.



Figure 1: Example of entry into an informal home

- Drill a 5mm pilot hole through the zinc into the trusses of the house.
- Use a pigtail screw and screw into the pilot hole.
- The hole needs to be sealed.
- Secure the drop cable by wrapping a dead-end onto the drop cable and hooking it to the pigtail screw.
- The drop fiber needs to make a small loop before entering the house to ensure the water drops off and not run into the house.
- Enter the house by drilling a hole and installing an electrical gland to secure the drop entry.
- The entry point should be sealed on the outside to prevent any weather damage.
- If the drop entry point is between the roof and the trussers, a 12/10mm duct needs to be used to protect the drop from heat and pinching. Duct to be sealed.



5.2 **Cable routing**

Fibertime™ has a high standard for neatness and all cable runs will follow the following criteria.

- Cable runs are to be as unintrusive as possible and thusly cable runs shall be limited to natural contours of the structure such as corners of walls, along skirting or ceilings.
- Cables runs in open space on walls shall only be utilised as a last resort.
- Cable runs within the house should be as short as possible.
- All cable slack from the installation should be neatly hidden behind the fibertime™ Optical Network Terminal (ONT) or the mounting bracket.
- The drop cable will be coiled one time in the slack management on the back of the Optical Network Terminal (ONT) before plugged into the PON port.
- The power supply cable of the Gizzu (UPS) must be secured to the wooden board and neatly routed to the main power supply of the house by using a glue gun or saddles along the skirtings.
- Cable to be installed to other surface that is not exposed zinc or surfaces that can be heat up exponentially.
- Saddles that are used must be in spec with the OD of the cable min 3mm ID min of 300mm spacing between saddles and use clear silicone to secure cable to surface.

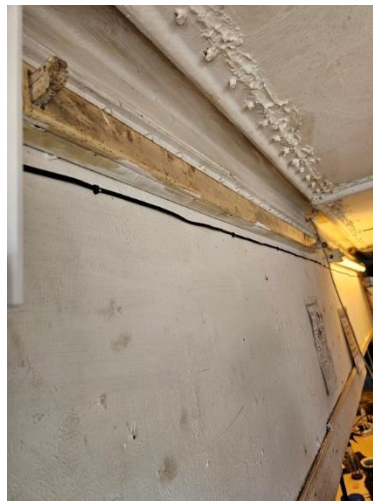


Figure 2: A neat cable run in an informal home

5.3 **ONT/mini-UPS setup and brackets**

The ONT/mini-UPS setup consists of the following;

1. Varnished Wooden Board (280mm x 120mm) with 4 pre-drilled holes.



2. ONT bracket.
3. X 2 flat head, self-tapping screws for the ONT bracket.
4. X 2 self-tapping screws or fisher plugs for bricks for the Varnished Wooden Board (280mm x 120mm).
5. X 1 black cable tie for mounting the Gizzu.
6. Nokia Optical Network Terminal (ONT).
7. Mini-UPS(Gizzu).
8. 10m 3 prong Power supply cord for the Gizzu.
9. Multi plug.
10. Fibertime™ ONT sticker.
11. Brady Tape to enter Drop Number that matches drop label at distribution point.

i. ONT/mini-UPS installation steps :

1. Varnished Wooden board (280mm x 120mm)

- The wooden board needs to be varnished and the edges need to be trimmed.
- Fix the wooden board against the wall by using a level. When level, mark the 2 holes and drill 2 holes into the wall with a 6mm bit.
- Tighten the wooden board to the wall using x 2 self-tapping screws or fisher pugs for bricks.



Figure 3: A varnished wooden board for ONT installation

2. ONT bracket

- Drill a 5mm hole in the middle of the ONT bracket for the cable tie to go through.
- Place the ONT bracket in the middle of the wooden board, mark the 2 holes and fix it to the board with x 2 flat head, self-tapping screws. Make sure a level has been used before the screws are tightened.





Figure 4: An ONT bracket installed in an informal home

3. Drop cable slack

- Coil the drop cable slack (no more than 10m) to a loop of no bigger than 300mm diameter. This slack is to prevent breaking the fibre when the ONT is unmounted for maintenance.
- Use 2x black cable ties on the loop to keep it neat.
- Put the drop cable slack over the ONT bracket before installing the mini-UPS.

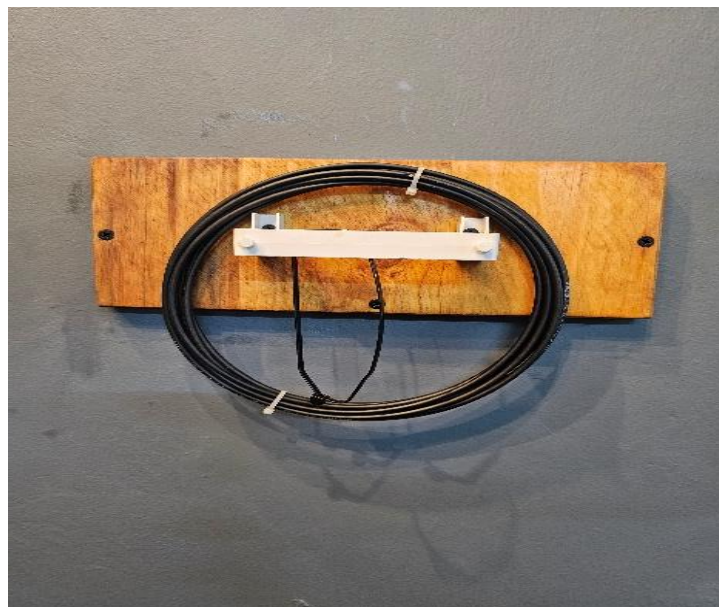


Figure 5: An indication of the placement of the drop cable slack

4. Mini-UPS (Gizzu) installation

- Remove the plastic tape that is on the middle of the Gizzu.
- Mount the mini-UPS power solution on top of the ONT bracket. Secure the



- mini-UPS by tightening it to the ONT bracket with a cable tie.
- Plug the supplied multi plug into the legal power supply of the house.
- Plug the AC power cable of the mini-UPS Power solution into the AC input socket of the house and the other end in the type D plug of the mini-UPS.
- Once this is done, and the extension cable is plugged in at the wall and switched on, the battery lights of the mini-UPS power solution will light up (this is not the case during loadshedding).
- Connect the DC output cable (2 ends) to the output socket of the mini-UPS power solution and the other end to the ONT DC jack port when installing the ONT.
- The output voltage selector on the mini-UPS power solution must be on 12V.
- Secure the power cable of the mini-UPS to the wooden board by using a glue gun or saddles.

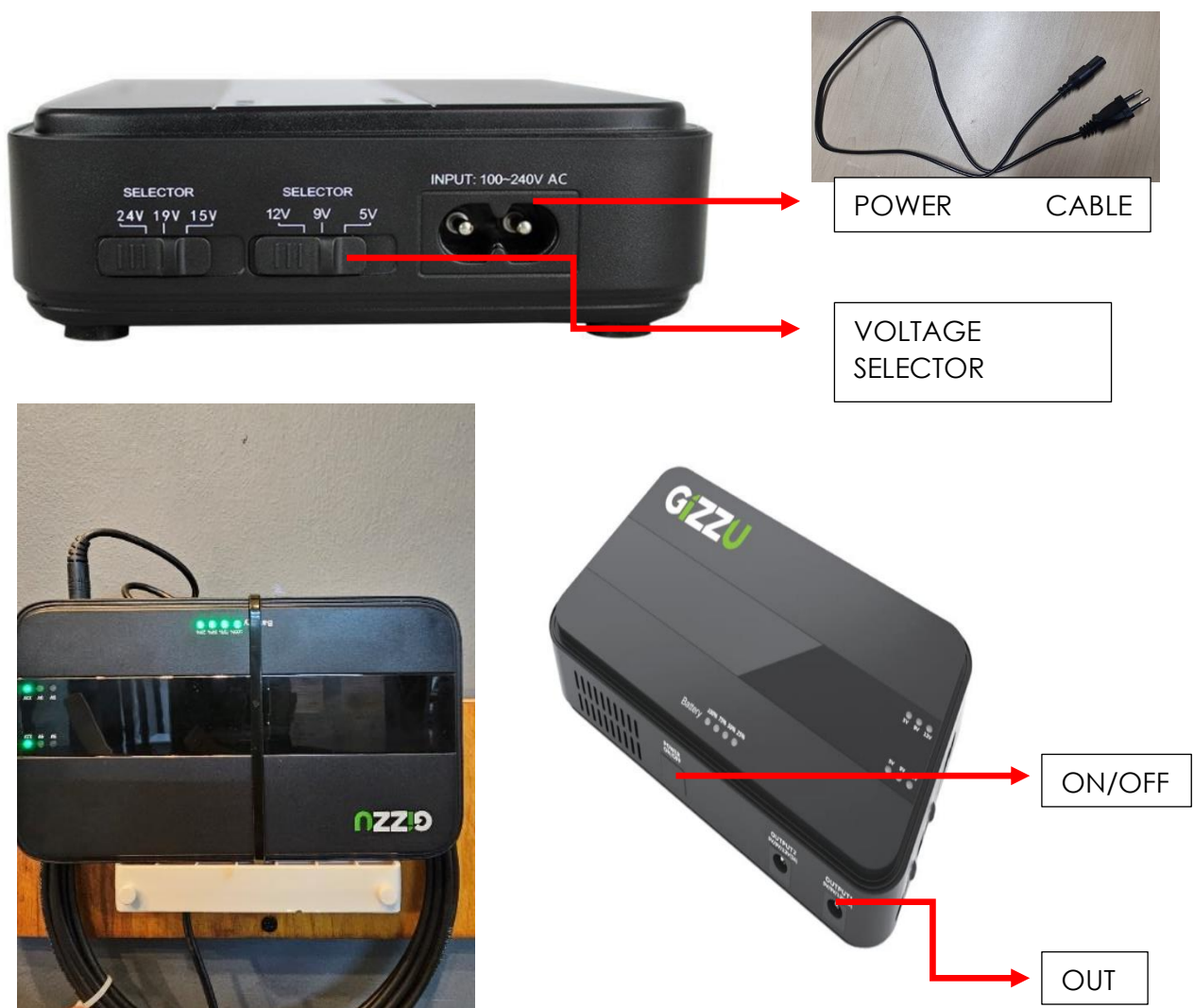


Figure 6: Schematics regarding the Mini-UPS (GIZU) installation



5. ONT installation

- Fiber shall be run vertically into the fiber entry point and looped at least once around the integrated fiber management before connecting into the PON port.
- Connect the DC output cable of the mini-UPS into the DC jack port of the ONT.
- Install the ONT onto the ONT bracket by sliding the ONT mounting holes into the ONT bracket.
- The antennas of the ONT must be in the upright position.
- Neaten the DC cable by using a small cable tie and store it behind the ONT.

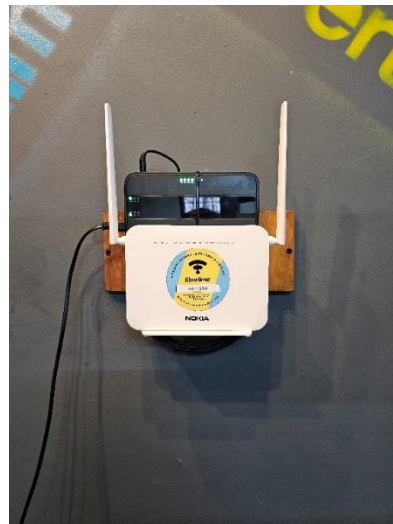
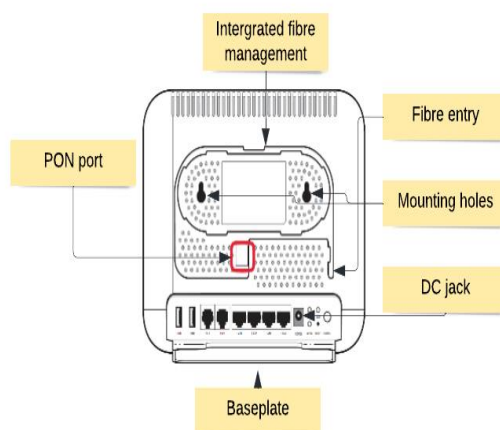


Figure 7: Schematics regarding the ONT installation

6. Fibertime™ sticker

- Apply the sticker onto the middle of the ONT. Make sure it is as straight and neat as possible and none of the lights are covered.
- Print the drop number according to Fibertime™ naming standards.
 - DRXXXX
- Apply the drop number label onto the sticker where the space is provided.
- The drop label must be black on yellow.
- Apply the small Serial number sticker of the ONT below the Nokia name.





Figure 8: Placement of the fibertime and Nokia stickers

7. Installed ONT/mini-UPS

- The final product needs to look like the attached photos.



Figure 9: Final installation of ONT and mini-UPS



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